1 Dimension

The next lemma is a key result which we will use repeatedly.

Lemma 1.1 (Plus/minus lemma)

Let S be a subset of a vector space V.

(a) If S is linearly independent and v is not in the span of S, then we can add v to S and S will still be linearly independent. That is, if S is independent, $v \in V - \text{span}(S)$, then $S \cup \{v\}$ is linearly independent.

(b) If $v \in V$ is in the span of the other vectors in S, then we can remove v from S without changing the span of S. That is, $\text{span}(S) = \text{span}(S - \{v\})$.

Exercise 1.2

Prove Theorem 1.1.

Proof:

1. Assume S is linearly independent and $v \notin \text{span}(S)$. We want to show that $S \cup \{v\}$ is independent. So, given $v_1, ..., v_n \in S$ and $\alpha_1, ..., \alpha_n, \alpha \in F$, we need to show if:

$$\alpha_1 v_1 + ... + \alpha_n v_n + v = 0,$$

then all the α_i, α are zero. There are two cases: $\alpha = 0$, $\alpha \neq 0$.

If $\alpha = 0$, then this equation reduces to:

$$\alpha_1 v_1 + ... + \alpha_n v_n = 0,$$

so by independence of S, all of these α_i are zero, so the α_i, α are zero and we are done.

If $\alpha \neq 0$, then we can rearrange the equation to get:

$$v = \frac{-\alpha_1}{\alpha} v_1 + ... + \frac{-\alpha_n}{\alpha} v_n,$$

so $v \in \text{span}(S)$, a contradiction of the assumption that v was not in the span. So, this case never occurs.

So, in all valid cases, the α_i, α are zero, showing independence.

2. It is trivial that $\text{span}(S - \{v\}) \subset \text{span}(S)$, since the latter space is the vector space of linear combinations of elements in S, while the former is the vector space of linear combinations of elements in a subset of S (and therefore all such combinations are also combinations of elements in S).

So, we need to check $\text{span}(S) \subset \text{span}(S - \{v\})$. Consider $u \in \text{span}(S)$. By definition, for some $\{v_i\} \in S$, we have scalars $\{\alpha_i\}$:

$$\sum_{i=1}^{n} \alpha_i v_i = u.$$

There are again two cases - either none of the v_i are equal to v, or one of them is. In the former case, this means all the $v_i \in S - \{v\}$, and so $\sum_{i=1}^{n} \alpha_i v_i \in \text{span}(S - \{v\})$ as desired.

So, we must treat the latter case. Without loss of generality, let $v_1 = v$, then we have:

$$\sum_{i=2}^{n-n} \alpha_i v_i + \alpha_1 v = u.$$

Since v is in the span of $S - \{v\}$ by assumption, there exist elements $v_{n+1}, ..., v_{n+k}$ not equal to v for some k and scalars $\alpha_{n+1}, ..., \alpha_{n+k}$ such that:

$$\sum_{i=n+1}^{n+k} \alpha_i v_k = v.$$

Substituting into the prior equation, we have:

$$\begin{aligned}
\sum_{i=2}^{n-n} \alpha_i v_i + \alpha_n v &= \sum_{i=2}^{n-n} \alpha_i v_i + \alpha_n \left(\sum_{i=n+1}^{n+k} \alpha_i v_i \right) \\
&= \sum_{i=1}^{n-1} \alpha_i v_i + \sum_{i=1}^{k} \left(\alpha_n \alpha_{n+i} \right) v_{n+i} \\
&= \sum_{j=1}^{n+k-1} \beta_j v'_j \\
&= u,
\end{aligned}$$

where $v_j = v_{i+1}$ and $\beta_j = \alpha_{j+1}$ for $j \in \{1, ..., n-1\}$, $\beta_j = \alpha_n \alpha_{j+1}$ for $j \in \{n, ..., n+k-1\}$.

This shows that u is a linear combination of $v_j \neq v$ and therefore is in $\text{span}(S - \{v\})$.
Notice that the notation here is kind of horrendous with lots of re-indexing, but all we are doing is writing u as a linear combination not equal to v plus a scalar times v, then rewriting v as a linear combination of elements not equal to v, then combining them.



Recall that a finite-dimensional vector space is a vector space with a finite spanning set.



Theorem 1.3

Let V be a finite-dimensional vector space and $S \subset V$. Then:

1. If $\text{span}(S) = V$ and S is finite, then S contains a subset B that is a basis of V.
2. If S is linearly independent, then S can be extended to a basis of V.



Proof:

1. Consider a set S with $\text{span}(S) = V$. If S is independent, we are done. If S is not independent, there exists some v that is a span of elements in $S - \{v\}$. By the prior lemma, we can remove

V and $\text{span}(S - \{v\})$ is still equal to V. Repeating this process (we can do this by finite-ness for finite and/or countability for S, or more generally via choice for infinite S), either this process will terminate or we will remove every element $v \in S$ from S while S still spans V, meaning that $\emptyset$ spans V. But, $\text{span}(\emptyset) = \{0\}$ since the empty sum is equal to the identity 0, so V has a basis $\{0\}$ and we are done (alternatively, we defined independence such that $\emptyset$ is independent).

2. Conversely, consider $S \subset V$ that is independent. Since V is finite-dimensional, we have some spanning set $S' = \{v_1, ..., v_n\}$. Either $\text{span}(S) = V$, or there exists some $v \in V$ such that $v \notin \text{span}(V)$.

   We claim the following:

   Lemma 1.4

   There is at least one $v_k \in S'$ not in $\text{span}(S)$.

   Proof:

   Suppose all the $v_k \in \text{span}(S)$. Then, the v_k are linear combinations of elements of S, that is, $v_k = \sum_{j=1}^{m} \alpha_{m_j} u_j$ for $u_1, ..., u_m$ in S for all v_k.

   Consider the v as mentioned above such that $v \notin \text{span}(V)$. Since the v_k span V, we have for some $\{\beta_i\}$:

$$\begin{aligned} v &= \sum_{i=1}^{n} \beta_i v_i \\ &= \sum_{i=1}^{n} \beta_i \left(\sum_{j=1}^{m} \alpha_{m_j} u_j \right) \\ &= \sum_{j=1}^{m} \left(\sum_{i=1}^{m} \beta_i \alpha_{m_j} \right) u_j, \end{aligned}$$

   showing that v is a linear combination of the u_j and is therefore in the span of S, a contradiction. Therefore, at least one v_k must not be in $\text{span}(S)$

   Therefore, by theorem 1.1, the set $S \cup \{v_k\}$ is still independent. If it does not span V, we again must have some v_{k_2} that is not in $S \cup \{v_k\}$, so we can append v_{k_2}. We continue to do this until we find a spanning set that is independent (at which point we have a basis), or we append all the v_k, at which point we are guaranteed to span V (since the v_k alone span V). Since we have reached this point while maintaining independence, this is also independent, and so we have a basis.

Corollary 1.5

Every finite-dimensional vector space has a finite basis.

Exercise 1.6

Prove theorem 1.5.

Proof:

By definition, a finite-dimensional vector space has a finite spanning set. Apply theorem 1.3 to reduce this finite spanning set to a basis.

Theorem 1.7

Let V be a finite-dimensional vector space which is spanned by a set of m vectors $\{v_1, ..., v_m\}$. Then, any linearly independent set of vectors in V is finite and contains no more than m elements.

Proof:

To prove the theorem it suffices to show that every subset of V which contains more than m vectors is linearly dependent. Let S be such a subset. Then S contains distinct vectors $w_1, ..., w_n$ with $n > m$. Since $\{v_1, ..., v_n\}$ spans v, we can write the w_i by:

$$w_i = \sum_{k=1}^{m} \beta_{ik} v_k.$$

Suppose there are scalars $\alpha_i \in F$ such that:

$$\sum_{i=1}^{n} \alpha_i w_i = 0.$$

Substituting in our prior equation gives:

$$\begin{aligned} \sum_{i=1}^{n} \alpha_i \left(\sum_{k=1}^{m} \beta_{ik} v_k \right) v_k &= \sum_{i=1}^{n} \sum_{i=1}^{m} \alpha_i \beta_{ik} v_k \\ &= \sum_{i=1}^{m} \left(\sum_{i=1}^{n} \alpha_i \beta_{ik} \right) v_k = 0, \end{aligned}$$

where we have used the fact that finite sums can have their order swapped and arithmetical properties in V. We want to force all the α_i to be zero to show independence of the w_i.

One way to get zero is by letting all the coefficients $\sum_{i=1}^{n} \alpha_i \beta_{ik} = 0$. If W is independent, there cannot exist nontrivial α_i satisfying this system, which can be realized as the following in matrix form:

$$\begin{bmatrix} \beta_{11} & \beta_{21} & ... & \beta_{n1} \\ \beta_{12} & \beta_{22} & ... & \beta_{n2} \\ \vdots & \vdots & & \vdots \\ \beta_{1m} & \beta_{2m} & ... & \beta nm \end{bmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

However, this is a homogenous system of m equations in n unknowns with more unknowns than unknowns. We will find feee variables when solving this system and therefore some nontrivial solution for the α_i. Therefore, the w_i cannot be independent (since this forces the α_i to be trivial) and this shows the w_i are therefore dependent, as desired.

Here is an alternate argument (see Dr. Trimm's De Rham Cohomology notes). I think this follows Axler's ordered tuples approach, which is kind of weird?:

Proof:

Suppose $\text{span}(\{v_1, ..., v_n\}) = V$. To prove the lemma, it suffices to show that every subset of V with $n+1$ vectors is linearly dependent. Let $\{w_1, ..., w_{n+1}\}$ be such a set. If $w_i = 0$ for some i, this is dependent, so we can say all w_i are not zero.

Since $w_i \in (v_1, ..., v_n\} = V$, we see that the set $\{w_1, v_1, ..., v_n\}$ is dependent (w_1 is a nontrivial combination of other vectors). Keeping an ordering $(w_1, ..., w_n)$, we can rewrite the linear combination that makes w_i a linear combination of the $v_1, ..., v_n$ to identify some v_j that is a linear combination of w_i (at least one of the v_i has a nontrivial coefficient in this expansion as otherwise w_i would be zero) and the other v_i, so we remove this from the spanning set to see that $\{w_1, v_1, ..., v_{j-1}, v_{j+1}, ..., v_n\}$ still spans V.

Now, we will attempt induction on the number of vectors replaced by w_i to show that we can replace every element with w_i (we have just shown the base case). Assume by the inductive hypothesis we have a set $\{w_1, ..., w_{k-1}, v_k, ..., v_n\}$ spanning V. Like before, we saw that $(w_1, w_2, ..., w_k, v_k, ..., v_n)$ is dependent, so one of these can be written as a linear combination of the preceding ones be label (i.e only the vectors before u in the tuple can be part of the combination). Some re-ordering of the w_i and v_k may be necessary to achieve this, but it is unimportant. If this is one of the w_i, this means that the w_i are dependent. Otherwise, we can again take out one of the v_i, and the reduced set spans V yet again. This shows the inductive step.

Therefore, we can replace vectors until either we obtain that the w_i are dependent (via the w_i being the vector that causes dependence in the prior paragraph), or that $\{w_1, ..., w_n\}$ replaces all the v_i and still spans V. If the w_i are not already dependent by now, appending w_{n+1} shows this is dependent (since the $w_1, ..., w_n$ span V and therefore w_{n+1} is a combination of them).

Corollary 1.8

Every basis of a finite-dimensional vector space is finite.

Exercise 1.9

Use Theorem 1.7 to prove **??**.

Proof:

We know there is at least one finite basis by Theorem 1.5, so there is a set of cardinality m of linearly independent vectors. Then, by Theorem 1.7, any set of independent vectors has cardinality $\leq m$, and in particular is finite. Since bases are linearly independent, this shows the result.

Corollary 1.10

If V is a finite-dimensional vector space, then any two bases of V have the same number of elements.

Exercise 1.11

Prove Corollary 1.13.

Proof:

Suppose $\mathcal{B}_1$, $\mathcal{B}_2$ are two bases of V with m and n elements respectively (we know these are finite by theorem 1.8). Then, since these are both independent sets, we know $m \leq n$ and $n \geq m$ - otherwise, $\mathcal{B}_1$ would be an independent set with more vectors than the independent set $\mathcal{B}_2$ and vice versa, which we know is impossible by Theorem 1.7. So, we have $m \leq n \leq m$, so we must have $n = m$.

Definition 1.12 (Definition)

Let V be a finite-dimensional vector space. The number of elements in any basis of V is called the *dimension* of V, and is denoted by $\dim V$.

Example 1.13

(a) $\emptyset$ is a basis for $\{0\}$, so $\dim\{0\} = 0$.

(b) The set $\{e_i\}_{i=1}^n$ (where $e_i \in F^n$ has a 1 in the i-th entry and 0s elsewhere) is a basis for F^n, so $\dim F^n = n$.

Corollary 1.14

Let V be a finite-dimensional vector space and let $n = \dim V$. Then:

(a) Any subset of V which contains more than n vectors is linearly dependent.

(b) No subset of V which contains fewer than n vectors can span V.

Proof:

(a) We have a basis of size n, so we have an linearly independent set containing n vectors, and the result follows from Theorem 1.7.

(b) If we have such a set S with $|S| < n$, by Theorem 1.3, we can reduce S to a smaller set S' such that S' is a basis with $|S'| \leq |S| < n$, implying that we have two bases with different sizes, a contradiction of the fact that dimension is well-defined.

Corollary 1.15

Let V be a finite dimensional vector space with $\dim V = n$. Then:

(a) Any set of n linearly independent vectors of V is a basis of V.

(b) Any set of n vectors which spans V is a basis of V.

Exercise 1.16

Prove Corollary 1.19.

Proof:

1. By Theorem 1.3, we can extend this set SB of n independent vectors to a basis B' with $|B'| \geq |B| = n$. But, all bases have the same size, so we see $n = |B|' \geq |B| = n$, so the only possibility is $|B|' = |B|$, and since $B \subset B'$ and both sets are finite, this implies $B = B'$, so B is already a basis as desired.

2. Essentially, apply the same argument in reverse. A spanning set S can be reduced to a basis S' with $|S'| \leq |S| = n$. But, all bases have the same number of elements, so we have $n = |S|' \leq |S| = n$, so $|S'| = |S|$, and since $S' \subset S$ and both sets are finite, this shows $S' = S$.





Example 1.17

Consider the set $S = \left\{ \begin{bmatrix} -5 \\ 3 \end{bmatrix}, \begin{bmatrix} 21 \\ 16 \end{bmatrix} \right\} \subset \mathbf{R}^2$. Since these two vectors are not scalar multiples of each other, S is linearly independent and hence a basis of $\mathbf{R}^2$ by the above theorem.